

Name: _____

Class: _____

Date: _____

Homework 3.1

Variables and Assignments

Score: _____ / 8

Multiple Choice

1. What is the value of b?

```
a ← 5  
b ← a  
a ← 8
```

- A. 13
- B. 8
- C. 5
- D. a

2. What is the value of x?

```
x ← 3  
x ← x * 2  
x ← x + 1
```

- A. 6
- B. 7
- C. 8
- D. 9

3. Which is the best variable name for storing the total cost of an order?

- A. x
- B. t1
- C. totalCost
- D. variable

4. What are the values after this code?

```
a ← 1  
b ← 2  
c ← a + b  
a ← c
```

- A. a = 1, b = 2, c = 3
- B. a = 3, b = 2, c = 3
- C. a = 3, b = 3, c = 3
- D. a = 2, b = 1, c = 3

Written 1

Complete a trace table (value of each variable after each line):

```
m ← 7  
n ← m - 2  
m ← m + n  
n ← m MOD n
```

Line	m	n
1	7	
2		
3		
4		

Written 2

Write code that swaps the values of variables `left` and `right`. Explain why a third variable is needed.

Name: _____

Class: _____

Date: _____

Homework 3.2

Data Abstraction

Score: ____ / 8

Multiple Choice

1. `lst ← [4, 7, 1]`. What is `lst[2]`?

- A. 4
- B. Error
- C. 1
- D. 7

2. On the AP exam, what is the index of the first element in a list?

- A. 0
- B. It depends
- C. -1
- D. 1

3. What is a main benefit of using a list?

- A. It manages complexity by storing many related values under one name
- B. It makes the program run faster
- C. It encrypts data
- D. It removes the need for variables

4. `letters ← ["a", "b", "c"]`. What happens when `letters[4]` is accessed?

- A. An error occurs
- B. Displays "a"
- C. Displays "c"
- D. Displays nothing

Written 1

Rewrite this code using a list, and explain one benefit:

```
s1 ← 90  
s2 ← 85  
s3 ← 77  
DISPLAY((s1 + s2 + s3) / 3)
```

Written 2

Explain what "index out of bounds" means. Give an example with a 3-element list.

Name: _____

Class: _____

Date: _____

Homework 3.3

Mathematical Expressions

Score: ____ / 8

Multiple Choice

1. What is $17 \text{ MOD } 3$?

- A. 5.67
- B. 5
- C. 2
- D. 0

2. What is $20 \text{ MOD } 5$?

- A. 0
- B. 4
- C. 5
- D. 15

3. What is $3 + 4 * 2$?

- A. 14
- B. 11
- C. 10
- D. 9

4. **Sequencing** means...

- A. choosing between options
- B. code statements run in the order they are written
- C. repeating steps
- D. sorting data

Written 1

Evaluate: (a) $29 \text{ MOD } 6$ (b) $3 \text{ MOD } 10$ (c) $8 + 12 / 4$ (d) $(8 + 12) \text{ MOD } 7$

Written 2

Write a Boolean expression that is true when `year` is divisible by 4 but NOT by 100.

Name: _____

Class: _____

Date: _____

Homework 3.4

Strings

Score: ____ / 8

Multiple Choice

1. What does `concat("snow", "man")` return?

- A. "snow man"
- B. "mansnow"
- C. "snowman"
- D. "snow"

2. How many characters are in "hi there" ?

- A. 7
- B. 8
- C. 2
- D. 9

3. A **substring** is...

- A. a number
- B. part of an existing string
- C. a list of strings
- D. an empty string

4. Which of these is a **string**?

- A. 42
- B. "42"
- C. true
- D. [4, 2]

Written 1

Write ONE expression using `concat` and `substring` that creates a username from `first` and `last`: the first 3 letters of `first` + the full last name. ("Taylor", "Kim" → "TayKim")

Written 2

Trace and give the final value of `out` :

```
out ← ""
FOR EACH w IN ["go", "for", "it"]
{ out ← concat(out, substring(w, 1, 1))
}
```

Name: _____

Class: _____

Date: _____

Homework 3.5

Boolean Expressions

Score: ____ / 8

Multiple Choice

1. What does $(4 > 2) \text{ AND } (3 = 3)$ evaluate to?

- A. 4
- B. false
- C. true
- D. Error

2. What does $\text{NOT } (5 < 3)$ evaluate to?

- A. true
- B. false
- C. 5
- D. 3

3. What does $(2 > 5) \text{ OR } (1 = 1)$ evaluate to?

- A. 2
- B. false
- C. Error
- D. true

4. Which tests whether x is between 1 and 100, inclusive?

- A. $x \geq 1 \text{ AND } x \leq 100$
- B. $x \geq 1 \text{ OR } x \leq 100$
- C. $x > 1 \text{ AND } x < 100$
- D. $x = 1 \text{ AND } x = 100$

Written 1

Complete a truth table for $\text{NOT } a \text{ OR } b$ for all four combinations of a and b.

a	b	$\text{NOT } a \text{ OR } b$

Written 2

Translate to a Boolean expression: "The game is over if lives is 0 or time is less than 1, unless the player has a shield."

Name: _____

Class: _____

Date: _____

Homework 3.6

Conditionals

Score: ____ / 8

Multiple Choice

1. What is displayed?

```
x ← 7
IF (x > 5)
{ DISPLAY("hi") }
```

- A. hi
- B. Nothing
- C. 7
- D. x

2. In an IF/ELSE statement, how many of the two blocks run?

- A. Neither
- B. Both
- C. Exactly one
- D. It depends on the computer

3. In an IF statement with no ELSE, what happens when the condition is false?

- A. The block is skipped
- B. The block runs anyway
- C. An error occurs
- D. The program stops

4. What is displayed?

```
x ← 3
IF (x MOD 2 = 0)
{ DISPLAY("even") }
ELSE
{ DISPLAY("odd") }
```

- A. even
- B. even odd
- C. 3
- D. odd

Written 1

Write code that sets `shipping` to 0 if `total` is at least 50, and to 7 otherwise.

Written 2

Explain why this code can display TWO messages, and rewrite it so only one ever displays:

```
IF (n > 10) { DISPLAY("big") }
IF (n > 5) { DISPLAY("medium") }
```

Homework 3.7

Nested Conditionals

Score: ____ / 8

Multiple Choice

1. What is displayed?

```
x ← 15
IF (x > 10)
{
  IF (x > 20) { DISPLAY("A") }
  ELSE { DISPLAY("B") }
}
ELSE { DISPLAY("C") }
```

- A. A
- B. C
- C. B
- D. B C

2. A nested conditional is...

- A. a loop inside a loop
- B. a conditional inside another conditional
- C. a list inside a list
- D. two separate IF statements

3. Which is equivalent to the code below (no ELSE branches)?

```
IF (a)
{
  IF (b) { DISPLAY("X") }
}
```

- A. IF (a OR b)
- B. IF (NOT a)
- C. IF (a AND b)
- D. IF (b)

4. When writing a grade calculator with nested IF/ELSE, which condition should be checked first?

- A. score $\geq$ 90
- B. score $\geq$ 70
- C. score $\geq$ 60
- D. It doesn't matter

Written 1

Shipping rules: orders over 5 lbs cost \$12. Otherwise, express orders cost \$9 and regular orders cost \$4. Write nested conditionals using `weight` and `isExpress` to set `cost`.

Written 2

Explain why the order of conditions matters in a nested grade calculator. Use an example score.

Name: _____

Class: _____

Date: _____

Homework 3.8

Iteration

Score: ____ / 8

Multiple Choice

1. What is displayed?

```
REPEAT 3 TIMES  
{ DISPLAY("hi") }
```

- A. hi
- B. hi hi
- C. Nothing
- D. hi hi hi

2. When does a REPEAT UNTIL loop stop?

- A. After 10 times
- B. When the condition becomes false
- C. When the condition becomes true
- D. Never

3. What is x after this code?

```
x ← 0  
REPEAT 5 TIMES  
{ x ← x + 2 }
```

- A. 10
- B. 7
- C. 5
- D. 12

4. Which loop is an infinite loop?

- A. $i \leftarrow 1$, REPEAT UNTIL $(i > 5)$ { $i \leftarrow i + 1$ }
- B. $i \leftarrow 5$, REPEAT UNTIL $(i = 0)$ { $i \leftarrow i - 1$ }
- C. $i \leftarrow 1$, REPEAT UNTIL $(i = 0)$ { $i \leftarrow i + 1$ }
- D. $i \leftarrow 0$, REPEAT 3 TIMES { $i \leftarrow i + 1$ }

Written 1

Write code using REPEAT UNTIL that displays 5, 10, 15, ..., 50.

Written 2

This loop should count down from 10 by 3s and stop. Explain why it never stops, and fix it.

```
k ← 10  
REPEAT UNTIL (k = 0)  
{ k ← k - 3 }
```

Name: _____

Class: _____

Date: _____

Homework 3.9

Developing Algorithms

Score: _____ / 8

Multiple Choice

1. A robot faces north. It runs ROTATE_RIGHT() twice. Which way does it face?

- A. North
- B. East
- C. South
- D. West

2. What is a benefit of using existing, correct algorithms as building blocks?

- A. It guarantees no errors
- B. It removes the need for input
- C. It makes programs larger
- D. It reduces development time and testing

3. Which correctly describes an algorithm to find the average of a list?

- A. Divide the first element by the length
- B. Multiply all elements
- C. Find the largest element
- D. Add all elements, then divide by the number of elements

4. Two algorithms are **equivalent** if they...

- A. have the same number of lines
- B. were written by the same person
- C. use the same variable names
- D. produce the same output for every input

Written 1

Describe in plain English, then write in pseudocode, an algorithm that counts how many numbers in `list` are greater than the list's average.

Written 2

Give two different algorithms that set `m` to the larger of `a` and `b`. Explain how you'd test that they are equivalent.

Name: _____

Class: _____

Date: _____

Homework 3.10

Lists

Score: ____ / 8

Multiple Choice

1. `lst ← [1, 2]`. What is `lst` after `APPEND(lst, 5)`?

- A. `[5, 1, 2]`
- B. `[5]`
- C. `[1, 5, 2]`
- D. `[1, 2, 5]`

2. `lst ← ["a", "b", "c"]`. What is `lst` after `INSERT(lst, 2, "x")`?

- A. `["a", "b", "x", "c"]`
- B. `["x", "a", "b", "c"]`
- C. `["a", "x", "b", "c"]`
- D. `["a", "x", "c"]`

3. `lst ← [4, 5, 6]`. What is `lst` after `REMOVE(lst, 1)`?

- A. `[4, 5]`
- B. `[4, 6]`
- C. `[5, 6]`
- D. `[6]`

4. What is `LENGTH([3, 3, 3, 3])`?

- A. 1
- B. 3
- C. 12
- D. 4

Written 1

`len(str)` returns the number of characters in `str`. Write code that creates a new list `longWords` containing only the words in `words` with more than 4 letters.

Written 2

Write the list after each line:

```
q ← ["x", "y"]
INSERT(q, 1, "w")
APPEND(q, "z")
REMOVE(q, 2)
q[2] ← "v"
```

Name: _____

Class: _____

Date: _____

Homework 3.11

Binary Search

Score: ____ / 8

Multiple Choice

1. What must be true about data before binary search can be used?

- A. It must be sorted
- B. It must contain only numbers
- C. It must have an even length
- D. It must be small

2. Each step of binary search eliminates about...

- A. one element
- B. a quarter of the elements
- C. half of the remaining elements
- D. all but one element

3. Linear search can be used on...

- A. only sorted lists
- B. sorted or unsorted lists
- C. only unsorted lists
- D. only lists of numbers

4. What is the maximum number of elements binary search checks in a sorted list of 16 elements?

- A. 5
- B. 4
- C. 8
- D. 16

Written 1

Trace binary search for 33 in [4, 9, 15, 21, 28, 33, 40, 47] (8 elements; use the left middle). List each value checked.

Written 2

A list of 50 names is unsorted and will be searched just once. Should you sort it and use binary search? Explain.

Name: _____

Class: _____

Date: _____

Homework 3.12

Calling Procedures

Score: ____ / 8

Multiple Choice

1. When calling `area(3, 7)`, the values 3 and 7 are...

- A. parameters
- B. events
- C. return values
- D. arguments

2. What does `RETURN` do?

- A. Restarts the program
- B. Sends a value back to the caller and ends the procedure
- C. Displays a value
- D. Repeats the procedure

3. What is `add(3, 4)` ?

```
PROCEDURE add(a, b)
{ RETURN (a + b) }
```

- A. 34
- B. $a + b$
- C. 12
- D. 7

4. Parameters are...

- A. the actual values passed in a call
- B. outputs of a procedure
- C. input variables listed in the procedure definition
- D. comments

Written 1

Identify the parameters and arguments, then give the output:

```
PROCEDURE greet(name, times)
{
  REPEAT times TIMES
  { DISPLAY(name) }
}
greet("Bo", 2)
```

Written 2

`PROCEDURE tax(price)` returns the tax on a price. Write ONE statement that stores the total cost (price plus tax) of a \$40 item in `total`.

Name: _____

Class: _____

Date: _____

Homework 3.13

Developing Procedures

Score: ____ / 8

Multiple Choice

1. Procedural abstraction lets a programmer...

- A. write all code in one block
- B. delete procedures
- C. avoid using parameters
- D. use a procedure knowing what it does without knowing how

2. Modularity means...

- A. compressing code
- B. making a program one long block
- C. using only global variables
- D. dividing a program into separate subprograms

3. Adding parameters to a procedure makes it...

- A. generalized, so it can be reused with different inputs
- B. slower
- C. unable to return values
- D. harder to call

4. Code that is copied and pasted several times with small changes suggests that you should...

- A. add comments
- B. use more variables
- C. delete the code
- D. write a procedure with parameters

Written 1

Write a procedure `letterGrade(score)` that returns "A" for 90+, "B" for 80-89, "C" for 70-79, and "F" otherwise.

Written 2

Explain two ways that writing procedures helps manage complexity in a large program.

Name: _____

Class: _____

Date: _____

Homework 3.14

Libraries

Score: _____ / 8

Multiple Choice

1. A software library is...

- A. a collection of procedures that can be used in new programs
- B. a list of books
- C. a type of loop
- D. a data compression method

2. An API is...

- A. a programming language
- B. a specification for how the procedures in a library behave and can be used
- C. a computer virus
- D. a type of variable

3. Why is documentation important when using a library?

- A. It makes the code run
- B. It is required to compile
- C. It explains the behaviors provided and how to use them
- D. It encrypts the library

4. What is a main benefit of using libraries?

- A. They make programs larger
- B. They replace testing
- C. They remove all bugs
- D. They simplify the task of creating complex programs

Written 1

Documentation: `translate(text, toLang)` returns text translated into the language code `toLang`. Write a statement storing the Spanish ("es") translation of "good morning" in `greeting`. What did you NOT need to know?

Written 2

Give one advantage and one disadvantage of using an external library in your Create Task program.

Name: _____

Class: _____

Date: _____

Homework 3.15

Random Values

Score: ____ / 8

Multiple Choice

1. Which values can `RANDOM(1, 4)` return?

- A. 1, 2, 3
- B. Any decimal from 1 to 4
- C. 0, 1, 2, 3, 4
- D. 1, 2, 3, 4

2. How many different values can `RANDOM(0, 9)` produce?

- A. 9
- B. 0
- C. 11
- D. 10

3. What is the probability that `RANDOM(1, 5)` = 5?

- A. 1/4
- B. 5
- C. 1/2
- D. 1/5

4. Which value can `RANDOM(2, 6)` NOT produce?

- A. 2
- B. 7
- C. 6
- D. 4

Written 1

List every possible output:

```
x ← RANDOM(1, 2)
y ← RANDOM(3, 4)
DISPLAY(x * y)
```

Written 2

Write code that displays "rare item" with a 1-in-20 chance and "common item" otherwise.

Name: _____

Class: _____

Date: _____

Homework 3.16

Simulations

Score: ____ / 8

Multiple Choice

1. A simulation is...

- A. a list of random numbers
- B. an exact copy of reality
- C. a type of compression
- D. an abstraction of a more complex real-world object or phenomenon

2. What is a main benefit of simulations?

- A. They replace all experiments
- B. They are always 100% accurate
- C. They never use computers
- D. They let us investigate phenomena without real-world cost or danger

3. Why are random numbers often used in simulations?

- A. To avoid testing
- B. To make code shorter
- C. To model the variability of the real world
- D. To encrypt results

4. Removing details to simplify a simulation can...

- A. make results perfect
- B. introduce bias or inaccuracy
- C. remove all randomness
- D. make it run forever

Written 1

Design a simple simulation to test whether adding a second cafeteria line would reduce wait time. Name two values you would vary and one simplification you'd make.

Written 2

Write pseudocode that simulates rolling a die 600 times and counts how many 6s appear. About how many 6s do you expect?

Name: _____

Class: _____

Date: _____

Homework 3.17

Algorithmic Efficiency

Score: _____ / 8

Multiple Choice

1. Algorithmic efficiency is an estimate of...

- A. the screen size
- B. the number of programmers
- C. the length of variable names
- D. the computational resources (time, memory) an algorithm uses as input size grows

2. Which runs in a **reasonable** amount of time?

- A. None
- B. Exponential (2^n)
- C. Factorial ($n!$)
- D. Polynomial (like n^2)

3. Which runs in an **unreasonable** amount of time?

- A. n
- B. 2^n
- C. n^2
- D. n^3

4. A heuristic is...

- A. a sorting algorithm that is always optimal
- B. a type of loop
- C. a syntax error
- D. an approach that finds a good, not necessarily optimal, solution when an exact one is impractical

Written 1

Classify each as reasonable or unreasonable and explain: (a) checking each pair of 1,000 students (n^2), (b) trying every password made of n bits (2^n).

Written 2

A delivery app must plan routes for 300 stops. Explain why it uses a heuristic and what it gives up.

Name: _____

Class: _____

Date: _____

Homework 3.18

Undecidable Problems

Score: _____ / 8

Multiple Choice

1. An **undecidable** problem is one for which...

- A. the answer is always no
- B. no algorithm can always give a correct yes/no answer for all inputs
- C. the algorithm is slow
- D. there is no input

2. The Halting Problem asks whether...

- A. a number is prime
- B. a computer is turned off
- C. a list is sorted
- D. a program will finish running or run forever

3. Which is a **decidable** problem?

- A. Can any program be proven bug-free?
- B. Will any given program halt?
- C. Is this number prime?
- D. None of these

4. How is an undecidable problem different from a slow (unreasonable-time) problem?

- A. They are the same
- B. An undecidable problem can't be solved for all inputs no matter how much time is available
- C. Undecidable problems are faster
- D. Slow problems can never be solved

Written 1

Classify as decidable or undecidable and explain: (a) Is a list sorted? (b) Will a given program ever print the word "done" for a given input?

Written 2

Explain the difference between an undecidable problem and an unreasonable-time problem, with an example of each.
